

21	B	Since the length of the wire is connected into the circuit, the current passing through the two cross-sections R and S is the same, i.e. the rate of flow of charge is the same.
22	A	The $3\ \Omega$ and $6\ \Omega$ resistors are connected in parallel while both the $2\ \Omega$ resistors are connected in parallel. The two sets of parallel resistors are then connected in series with each other. Thus the current flowing into the two sets of parallel resistors is the same, though the individual current to the resistor may not be the same. Having the same p.d. across the first set of parallel resistors, the smaller resistance of $3\ \Omega$ allows a greater current to flow through. Conversely, having the same p.d. across the second set of parallel resistors, the equal resistance of $2\ \Omega$ allows the same current to flow through. Hence, $I_1 > I_2$ The effective resistance of the two sets of parallel resistors are $2\ \Omega$ and $1\ \Omega$ respectively. Thus, according to potential divider rule, $V_1 > V_2$
23	A	When the galvanometer has a current reading of zero, For minimum V_{output}, $R_{\text{output}} = 0\ \Omega$, $V_{\text{output}} = 0\ \text{V}$